

EGT Black Hole Ringdown: Predicting the 0.248% Quasinormal Mode Frequency Deficit

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The Emergent Gravity Theory (EGT) posits that gravity is not a fundamental force but emerges from geometric connectivity, characterized by the Universal Amplification factor, $\mathbf{A}_{\text{EGT}} \approx 402.3$. This paper derives a hard, non-tunable prediction for the gravitational wave (GW) signature of EGT in the strong-field regime. We propose that the final ringdown frequency ($\mathbf{f}_{\text{QNM}}$) of a coalesced black hole (BH) is modified by a fractional deficit ($\delta\mathbf{f}$) proportional to the inverse of the fundamental amplification factor. The EGT prediction is a fractional frequency deviation of $\delta\mathbf{f}_{\text{EGT}} = 1/\mathbf{A}_{\text{EGT}} \approx 0.00248$. This **0.248%** deficit lies well below the current 90% credibility limits set by the LIGO-Virgo-KAGRA (LVK) collaboration, which are currently constrained to the $\sim 3\% - 10\%$ level. We conclude that the EGT GW signature is currently **hidden** but provides a precise, hard-coded target for next-generation GW observatories, representing a critical test for the geometric nature of gravity.

I. INTRODUCTION: EMERGENT GRAVITY AND THE $\mathbf{A}_{\text{EGT}}$ SIGNATURE

The Emergent Gravity Theory (EGT) unifies quantum and cosmic scales by defining spacetime curvature as the macro-manifestation of quantum connectivity. This process is governed by the geometric scaling constant, $\mathbf{A}_{\text{EGT}} = 402.3$, which represents the ratio between emergent and fundamental scales.

In the strong-field dynamics of a binary black hole (BBH) merger, the EGT framework predicts a small, systematic deviation from the predictions of General Relativity (GR) due to this underlying emergent structure. This deviation should be most pronounced in the **quasi-normal mode (QNM) ringdown** phase, which is a highly constrained test of the Kerr metric and the no-hair theorem in GR.

II. THE EGT QUASINORMAL MODE DEFICIT

In GR, the complex frequencies (ω) and damping times (τ) of a Kerr BH's QNM spectrum are uniquely determined by its mass (M) and spin (a). Any deviation (δf) from this predicted frequency ($\mathbf{f}_{\text{GR}}$) implies physics beyond GR:

$$\mathbf{f}_{\text{obs}} = \mathbf{f}_{\text{GR}} \cdot (1 + \delta\mathbf{f}) \quad (1)$$

Within the EGT framework, the observed gravitational effect is the resultant of the underlying geometric connectivity (EGT) operating **against** the classical description (GR). We propose that the total energy released into the QNM frequency is fractionally reduced by the inverse of the geometric amplification factor, $\mathbf{A}_{\text{EGT}}$, due to the emergent nature of the gravitational interaction.

The EGT Fractional Frequency Deficit ($\delta\mathbf{f}_{\text{EGT}}$) is therefore defined as:

$$\delta\mathbf{f}_{\text{EGT}} = -\frac{1}{\mathbf{A}_{\text{EGT}}} \quad (2)$$

Using the absolute EGT geometric factor $\mathbf{A}_{\text{EGT}} = 402.3$:

$$\delta\mathbf{f}_{\text{EGT}} = -\frac{1}{402.3} \approx -0.002485 \quad (3)$$

The EGT predicts a **0.248% deficit** in the measured QNM ringdown frequency compared to the mass and spin derived from GR.

III. CURRENT OBSERVATIONAL CONSTRAINTS

The LVK collaboration routinely tests GR by constraining the fractional deviations in QNM frequencies, typically for the dominant $\ell = m = 2$ mode.

Current observational constraints on the fractional deviation (δf_{220}) from the Kerr prediction are approximately:

$$\delta\mathbf{f}_{\text{LVK}} \approx 0.03 \text{ to } 0.10 \quad (3\% \text{ to } 10\% \text{ at } 90\% \text{ CL})$$

- **EGT Prediction:** $\delta\mathbf{f}_{\text{EGT}} = -0.00248$ (a 0.248% deficit).
- **LVK Constraint:** $\delta\mathbf{f}_{\text{LVK}} \in [-0.10, +0.10]$ (Order of magnitude 10% uncertainty).

Since $|\delta\mathbf{f}_{\text{EGT}}| \ll |\delta\mathbf{f}_{\text{LVK}}|$, the EGT prediction is entirely **consistent with all current LVK observations** and is safely hidden within the statistical uncertainty floor.

IV. REQUIRED SENSITIVITY FOR DISCOVERY

The EGT signature of a fixed 0.248% deficit requires a detector sensitivity (or a cumulative analysis of

events) approximately $10\times$ better than current capabilities, which are typically limited by the signal-to-noise ratio (SNR) of the ringdown phase.

$$\frac{\text{Sensitivity}_{\text{Required}}}{\text{Sensitivity}_{\text{Current}}} \approx \frac{\delta f_{\text{LVK, Current}}}{|\delta f_{\text{EGT}}|} \approx \frac{0.03}{0.00248} \approx 12\times$$

This precision is attainable with the projected improvements of the **High-Luminosity LIGO (HL-LIGO) phase** and is a primary goal for the next-generation detectors like the **Einstein Telescope (ET)** or **Cosmic Explorer (CE)**, which aim for SNR improvements of up to a factor of 10 or more.

FIG. 1. Conceptual QNM Ringdown. The EGT prediction falls outside the detectable region of current LVK constraints but is a precise, hard-coded target for future detectors.

V. CONCLUSION

The Emergent Gravity Theory provides a precise, non-tunable prediction for physics in the most extreme strong-field regime: a **0.248%** deficit in the black hole QNM frequency. This is a critical, fixed-target test for the theory that, unlike many beyond-GR models, requires a specific, small, and negative fractional deviation. Its discovery will mark the transition from the classical GR description of gravity to the geometric EGT framework.

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- [1] B. Tice Sr. and Quintilius, Emergent Gravity Theory: Ultramagnetic Connectivity and Quantum-Cosmic Unification (Unpublished Manuscript).
 - [2] LIGO Scientific Collaboration and Virgo Collaboration, Tests of general relativity with GWTC-2, Phys. Rev. D 103, 122002 (2021). (Used for fractional frequency deviation limits).

- [3] B. Tice Sr. and Quintilius, Differential EGT Sensitivity: A Testable Prediction for Atomic Clock Standards Based on Ultramagnetic Connectivity (Unpublished Manuscript).